



ESHA TANDON

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EDUCATION

B.E. Electrical Engineering	2023-2027	Netaji Subhas University of Technology	7.76 CGPA
CBSE (Class XII)	2023	Sadhu Vaswani International School for Girls	93.4%
CBSE (Class X)	2021	Sadhu Vaswani International School for Girls	97.8%

INTERNSHIP/S

SDE Intern, Stealth AI Startup (Remote)

May-26 to Jul-26

- Built **retrieval-augmented (RAG) workflows** for internal knowledge tools, improving answer grounding and cutting unsupported responses in testing.
- Shaped multi-tenant ingestion flows: document chunking, embeddings, and **secure tenant isolation** for reliable retrieval.
- Shipped frontend and API surfaces in React, FastAPI, and PostgreSQL, turning research ideas into production features.

ACADEMIC PROJECTS

SIRGOBLIN — AI Code Safety Platform | Python, MCP, CI/CD | GitHub

- Blocks unsafe AI-generated code before production, **scanning 100K lines in 22 seconds** across 52 rules from real outage postmortems.
- Built an **MCP server** exposing 7 verification tools over JSON-RPC, letting AI agents verify their own work mid-task.
- Cut AI review costs 4×** via model routing and batching; **raised detection accuracy 26%** auditing 8 open-source codebases.

INVARIANT — Adaptive DSA Practice Platform | Next.js, TypeScript, PostgreSQL | GitHub

- Tells users **which coding problem to solve next, and why**, replacing static problem lists across **32 DSA topics**.
- Built an **adaptive recommendation engine** blending **5 signals**, led by an Item Response Theory model predicting per-user solve probability.
- Added **multi-armed bandit (UCB) exploration** and **FSRS** memory-decay modeling to time reviews before forgetting.

NEXUS RAG — Multi-Tenant Agentic RAG Platform | FastAPI, pgvector, PostgreSQL | GitHub

- Multi-tenant agentic RAG platform with **bring-your-own-key sovereignty** for querying private document corpora.
- Designed autonomous retrieval using **hybrid pgvector and full-text search** for high recall over private data.
- Unified ingestion across **4 formats** (PDF, Markdown, PPTX, images) with chunking, embedding, and answer grounding.

ACADEMIC ACHIEVEMENTS AND AWARDS

- Secured **Top 200 nationally** in the Amazon ML Challenge, 2025.
- Qualified for **Round 3** of EY Techathon 2025 and **Round 3** of Smart India Hackathon 2025.
- Solved **400+** **data structures and algorithms problems** across LeetCode and other practice platforms.

TECHNICAL SKILLS

- Programming Languages** : C++ (Proficient), Python (Proficient), Java, JavaScript, TypeScript
- Web Technologies** : React, Next.js, HTML, CSS, shadcn/ui
- Backend & Databases** : FastAPI, REST APIs, PostgreSQL, Prisma, Supabase, pgvector, NextAuth, Docker
- GenAI & ML** : RAG, Agentic Workflows, Generative AI, MCP
- Core Subjects** : Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Signals & Systems

POSITIONS OF RESPONSIBILITY

- Member of **NSUT.AI**, the Artificial Intelligence & Machine Learning society of NSUT.

EXTRA-CURRICULAR ACTIVITIES AND INTERESTS

- Active competitive programmer on LeetCode and other practice platforms.
- Interests: Reading.